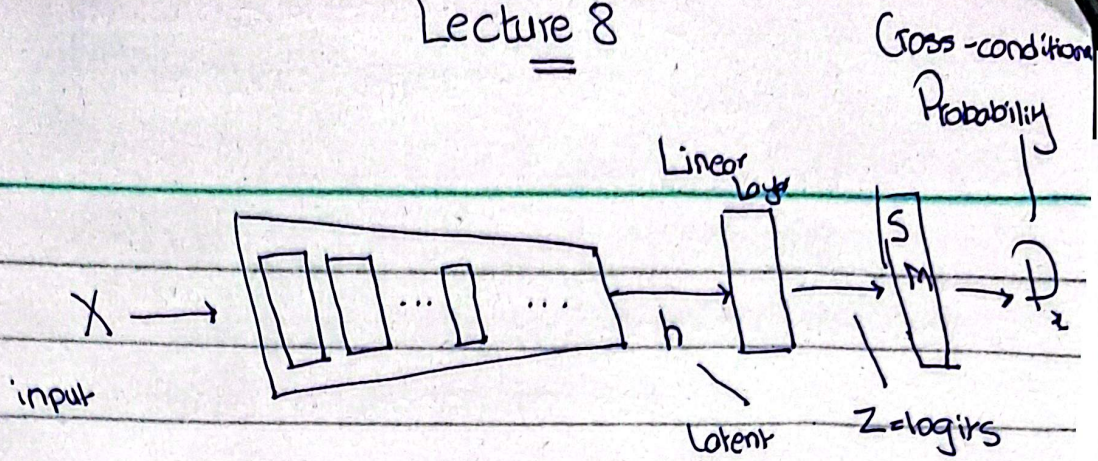


Lecture 8



Goal:

① Define an unknownness score $U(x)$

$$U(x) > \tau \rightarrow \text{reject}$$

$$U(x) \leq \tau \rightarrow \text{accept} \rightarrow \text{Detector}$$

τ - Threshold

② Classify x correctly - Classifier

Guiding principles:

1. Create some open-space for $x \notin Y_K$

$$2. P_U, P_K \Rightarrow U(x) = LRT = \frac{P_U(x)}{P_K(x)}$$

likelihood
ratio test

Softmax - relative - sum to 1

Z - Logits - absolute

Recognition Methods:

1. Post-hoc method:

train our model as usual
on images known (y_k)

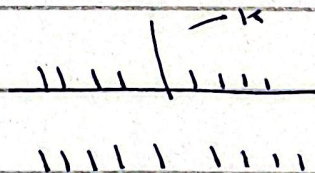
→ Unknowness Score measures:

MSP : Maximum
Softmax Probability

1. Unknowness score

$$U_{MSP}(x) = 1 - \max_k P_k$$

(Relative)



basically if model knows which class
is known based on ~~conf~~ confidence
(softmax)

↳ value of U_{MSP} will be ↓

↳ ~~value of U~~

Maximum Logit score

$$2. U_{MLS}(x) = - \max_k Z_k$$

(Absolute)

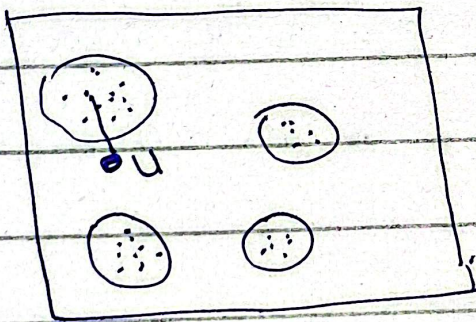
$$\rightarrow U_{\text{energy}}(x) = \log \sum_{k=1}^K e^{z_k} \quad \text{Covariance matrix}$$

— Mahalanobis distance

$$U_{\text{MAH}}(x) = \min_{c \in y_u} (h - u_c)^T \sum_{i=1}^{-1} (h - u_i)$$

From which class cluster
distance is minimum

Distance from every
class cluster



u_c — mean
 $\sum_{i=1}^{-1}$ — covariance matrix

basically the minimum distance to the nearest cluster

— if it is small enough, it is likely to be known

— if it's large, then it's will be unknown

best possible +ve augmentation
kia he

2022 — good closed set classifier
GCSE: Closed set
↓
basically a classifier is enough

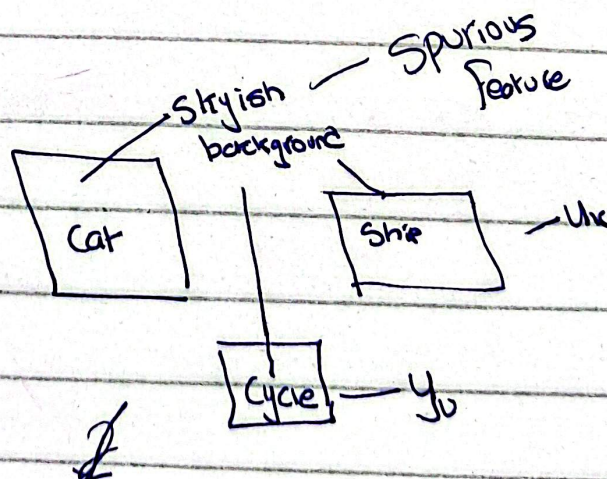
— if the model is able to understand clusters,
then its good enough

— if we take many images and not destroy the
catness of cat, is called positive augmentation

↓
→ if we train model on multiple textures of
cat, on all possible textures of cat, model
will ignore shape, and will think texture
matter
↳ robust

Problem:

example:

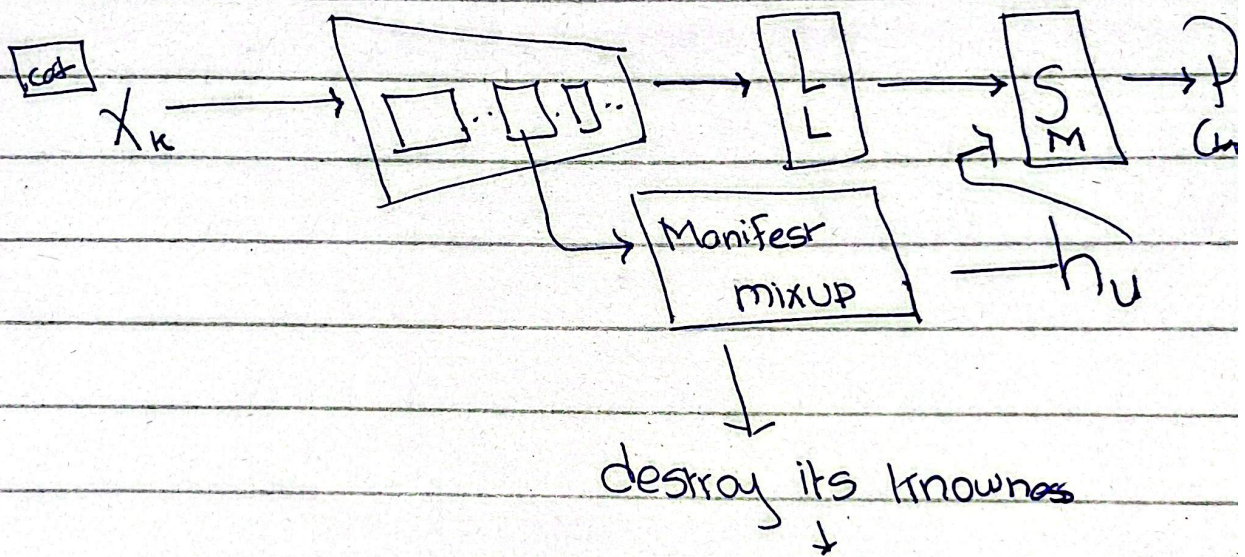
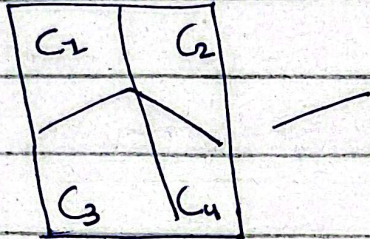


2-missing?

$C_1 - C_2 - C_3$

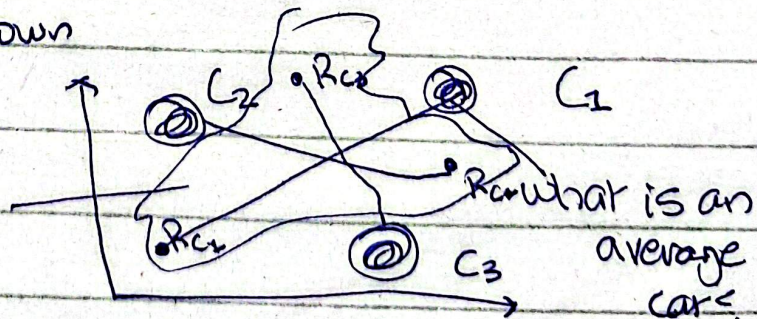
$C_1 - C_3$

③ Placeholder Learning (PROSER)



⇒ Take 2 classes and do linear interpolation that lies in center which is unknown

each 2 classes
create a
new class



RPL - Reciprocal point
Learning

- instead of learning what is cat,
learn what is not cat

- learn every class's reciprocal point



at that point, the
image is not cat.



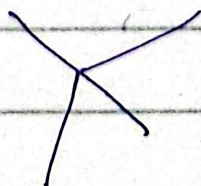
Thus, each class's reciprocal point
lies in open space

- Reciprocal point is not fixed,
we learn through gradient descent

* A reciprocal point of a class
will always be in open space
(no other class isn't in it)

Fail?

* geometrically
as clusters is separated by
lines



For every posthoc

- False confidence:

⇓

① Generic Feature Learning:

② Shared / Non-shared circuits:

① Generic Feature Learning:

└ pointed ears

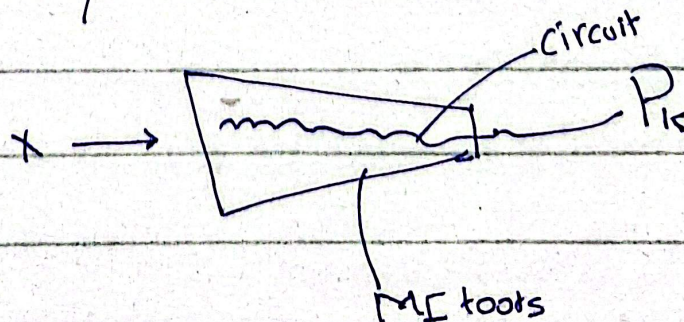
$h =$ └ Fur
└ Snout

$y_k = \{ \text{Dog}, \text{Truck} \}$

$y_u = \{ \text{wolf} \}$

When you have to focus only on discrimination,
nothing is stopping your model
to predict ~~your~~ wrong predictions

② Shared / Non-shared circuit



MI tools — LLM
↓
to view circuits

When we are giving an image to model for predicting, dog circuit will open when dog's image is added. What if wrong circuit opens if dog's image is added?

• Comparing OS models

↳ Measures of accuracy

CSA

	$U(x)$	Closed Set classifier
k_1	0.1	✓
k_2	0.2	✓
k_3	0.25	x
k_4	0.35	✓
k_5	0.5	✓
u_1	0.3	
u_2	0.45	
u_3	0.6	
u_4	0.8	
u_5	0.9	

1) CSA: closed set accuracy — $4/5 = 0.8$.

2) AUROC:

↓
Do unknowns generally receive larger $U(x)$?

= Probability that a random unknown looks more unknown than a random known

[25 pairs] - Known Sample → Compare this against all unknowns

② 0.1 → 0.3, 0.45, 0.6, 0.8, 0.9

⑤ 0.2 → 0.3, 0.4

⑤ 0.25

④ 0.35

③ 0.5

↓

$$= \frac{22}{25} = 0.88$$

↳ if 1 randomly select 1 unknown and 1 known then there is 88% chance that it will more be unknown than known